

In the Name of God

Comparing Two Methods of Teaching Calculus in University

Mohammad Hossein Pourkazemi

Associate Professor, Shahid Beheshti University

Abstract

Teaching calculus to non-mathematics students, like Economics, Management, Accounting, etc., is different from teaching the students with major in pure mathematics. In some universities, the lecturers teach pure theory and don't pay much attention to applications of the topics. Non-mathematics students should learn mathematics as a tool that they need to use it in their other courses, and they are not interested in the theory.

There are two methods of teaching calculus to non-mathematics students. In the first method only the pure theory is taught without giving any real life example of the applications. In the second method the emphasis is on the application rather than theory. The question is: which one of these methods will increase the motivation of students to learn mathematics? In this research we want to answer this question.

Although it is an important tool for economics and management analysis, most of the students are not very interested in mathematics,. Using my long experiences in teaching mathematics, I prove that giving applied examples in Economics and Management will create interests in mathematics among students. Also, by the help of an opinion poll, we observed that the students prefer the second method of teaching mathematics. Also we observed that students who have learned mathematics using the second method are more successful.

Introduction

There are 9 years of general schooling in Iran that consists of 5 years of elementary school, 3 years of guidance school, and the first year of high school. The students who want to continue their studies have two options; Vocational Studies or Theoretical Studies in the high schools that lead them to higher education at universities.

The theoretical studies have different branches. The students who are strong in mathematics usually continue their studies in mathematics and physics, those who are not interested in the mathematics study human sciences and literature, and the rest study experimental sciences.^[1]

After finishing high school, the students who want to enter higher education institutes study a one year pre-university courses in the related branches, i.e., mathematics and physics, experimental sciences, and human sciences and literatures. Then they take part in some university

entrance examination.

Usually, graduates of the mathematics and physics discipline study engineering and graduates of the experimental sciences study medical sciences, Dentistry or Pharmacology, and graduates of Human Sciences study law or political Sciences. This is the interest of around 80% of high school graduates.^[2]

Due to limited capacity of universities in the above fields, most of the students have to continue their studies in other branches, which they are not very interested in those subjects. Same is true for the students of economics and management need to know and use mathematics for their economic analysis and planning.

The main problem is lack of interest in mathematics among these students. The challenge is how we can encourage them to become interested in mathematics, and which teaching method is the best.

1) Economics and management student's lack of interest in mathematics

At the beginning, some of the students accepted in economics and management branches of the public universities are not interested in their field of studies. In 2005, around 1500 students were accepted in economics schools, and around 1300 students were accepted in the management sciences schools through 20 public universities in Iran.^[3]

Most of the students in economics and management branches, especially those who studied experimental and human sciences in high school, are not interested in mathematics. Some of them actually studied experimental and human sciences in high school because they did not like mathematics. Now they are studying economics and must study differential equations, Calculus I, and Calculus II courses compulsorily. So the first basic problem is, the knowledge of students in mathematics is not homogeneous. The second problem is that they mostly do not like mathematics at all. Operation research, Probability, and statistics are very important tools in teaching Macro and Micro economics. The challenge is “how we can encourage them to learn and like mathematics”?

2) Teaching mathematics in economics and management branches

There are two methods of teaching mathematics to the students of economics and management sciences branches in Iranian universities. In some universities they study mathematics together with the students of other branches, like Sociology, Management, and Biology, or the lecturers from the Departments of the mathematics come to their department and teach them mathematics. Most of these lecturers are not familiar with economics and management teaches mathematical theories from books like Thomas, *Calculus and Analytic Geometry*^[4] with less emphasis on proofs.

The second method is teaching mathematics by a mathematics lecturer who is familiar with economics or management ideas. The lecturer knows exactly what part of mathematics is

required for the undergraduate and graduate students in economics.^[5]

Teaching mathematics as a service course for other branches has its own problems that mathematics teachers are always interested in studying these problems.^[6] In ICME/8 conference held in Sevilla 14-21 July 1996^[7], this issue was discussed.^[8]

3) The Past Experience

About 30 years ago, I started my job as lecturer in the Faculty of Economics at Shahid Beheshti University. I have studied mathematics and, while studying for my master degree, I enrolled in an elective course in mathematical economics. I enjoyed the application of mathematics in economics.

I became interested in teaching mathematics for economic students. One of the lecturers had a general mathematics book that he called it “General mathematics for nonmathematical students”. He omitted difficult proofs of the book, but he was not familiar with economic concepts and so he was not interested in the application of mathematics in economics. This book was also thought to students of Management, Accounting, and Biology and so on.

I used this book in my classes too. The students who studied mathematics in high school did not have any problem for learning the book, but it was difficult for the other students. After a while I became responsible for teaching all mathematics courses in the Faculty of Economics. Each term around 250 students enrolled in this faculty to study Economics, Accounting, and Management.

After 3 to 4 years of teaching mathematics, I noticed that using applied examples encourages the students to pay more attention to the course. Especially, the first year students paid more attention to mathematics with applied examples. So I decided to find a book about applied mathematics and I found the book “Mathematical Analysis with Business and Economic Applications by: Jean E Weber” and I translated it into Persian.^[9]

I could not use this book as the main source of the course, because it was very easy for those with a good background in mathematics, since they have studied about 60 percent of the book in their high school ages, but the applied examples were interesting for them.

In 1988, I wrote a book called “General Mathematics and its Applications”, in which, after each chapter I gave examples of its application in Economics, Management, Physics, Mechanics, and Electronics. The application in Economics was separated from the application in other branches. This book is in two volumes and around 1200 pages divided into 11 chapters.

The contents of the book are: Set Theory, Complex Numbers, Functions, Derivatives, Application of Derivatives, Integral, Vectors and Matrices, Multi Variable Functions, Series, Differential Equations and Difference Equations. Each section of the book has many applications in economics. The book has three editions and been reprinted 18 times. It is used in all universities in Iran.^[10] I do not know of any similar experience in the other economic faculties in Iran.

Nowadays, high level mathematics is being used in economics, so the students should

know more and should be interested in the subject; otherwise they fail to pursue their studies in economics.

4) The Main Problem

The objective of this report is to show the author's effort to solve the mentioned problems in teaching mathematics to non-mathematics students. The success of the effort, regarding the students who did not like mathematics in high school, and were scared of the subject, is noticeable. The author was successful in creating interests in the students by changing the teaching method, and by this report asks from the other colleagues to use similar method in teaching mathematics to other non-mathematics students.

It is important to mention that, unlike some countries in which top students usually study economics, in Iran top students prefer to study engineering or medical sciences. In Human Sciences students prefer to study Law. So the students, who study economics in Iran, are those who don't like mathematics, and also are weak in other subjects in high school as well. Having influence in these conditions is very important and the big challenge is the slogan of "Mathematics for Every one". Researchers and the students of economics need to use advanced concepts of mathematics and statistics in their field of study. ^[5]

There are problems teaching subjects such as derivative integral to the non-mathematics students because of their non-homogeneous knowledge in mathematics and their lack of interest in this subject. The challenge is how to teach these students and make them like mathematics.

To solve the problem of student's non-homogeneous knowledge of mathematics, we offered a class for those students who did not study proper mathematics in high school. This class was compulsory for those who studied experimental sciences or human sciences in high school. Those who studied mathematics and physics in high school could also attend this course if they thought they need to improve their mathematics. After finishing this course, the mathematics knowledge of all of the students should be almost similar. This method is very effective in homogenizing the level of student's knowledge in mathematics.

Now, let us have a look at the main problem, i.e. creating interests in the students to learn mathematics. Students want to learn economics or management but, since they are not familiar with the topics we have to make them believe in the usefulness of mathematics as a necessary tool for their field of study. This can be shown to them with proper examples in each section. In my two volumes book, "General Mathematics and its Applications", in each section there are examples to show these applications. This book is in Persian and is based on the knowledge that the students learn in high school.

Now we want to see whether we were able to make the students interested in mathematics or not? What is the opinion of the students about this method? We were successful in our challenge, i.e. encouraging the students to learn mathematics with enthusiasm?

The other problem is; if there are two methods of teaching in Iran, we need to compare these two methods and see which one is more successful?

Last semester, I had two classes. One class was in Tehran University and the other class

was in Shahid Beheshti University. I decided to use each method of teaching in one of these classes, but the students did not let me continue teaching pure mathematics, which was an interesting issue. So I used the same method in both classes and I will compare two methods by other means that I will explain in the next sections.

5) The Questionnaire

At the end of the term, students of the both classes were asked to fill a questionnaire. In the questionnaire we asked them their type of high school diploma, and then we asked questions about their view points about these teaching methods and the applied examples. Also we asked them if they liked to learn mathematics taught to them by the lecturers from the Department of Mathematics who are not familiar with economics.

There were 500 students in the department and the number of sample was appropriate regarding the population. I have used a random sample of size 62 out of 500. The sample size was obtained in reference to the result of another opinion pool regarding the students' attitudes towards an economical problem in Iran. The questions and their answers are as follows:

Question (1): What is your high school diploma type?

There were 62 students in the sample from which 10 students studied mathematics and physics, 36 students studied Experimental Sciences, and 10 of them studied Human Sciences in high school.

Diploma	Mathematics	Experimental	Human Sciences	Total
Frequency	16	36	10	62
Per Cent	25.8	58.1	16.1	100

Question (2): Do you like to study mathematics with the students with majoring in mathematics?

The answers of the students to this question fall in 5 groups as follows:

Answer	Very Interested	Interested	No Differences	Not Interested	Not Interested At All
Frequency	0	0	2	18	42
Per Cent	0	0	3	29	68

Notice that close to 97% of the students were not interested to study mathematics in the Department of Mathematics. This is why they did not let me continue teaching them pure mathematics without applied examples.

Question (3): Do you like to study mathematics with the students of the other non-mathematics branches like Management, Chemistry, and Biology...?

The answers of the students to this question fall in 5 groups as follows:

Answer	Very Interested	Interested	No Differences	Not Interested	Not Interested At All
Frequency	0	4	10	14	34
Per Cent	0	6.5	16.1	22.6	54.8

Notice that most of the students are not interested in studying mathematics with the students from other branches.

Question (4): Were the applied examples explained for the Math I adequate?

The answers of the students to this question fall in 5 groups as follows:

Answer	Much More Required	More Required	It was Adequate	Less Required	Not Required At All
Frequency	12	14	22	10	4
Per Cent	19.4	22.6	35.5	16.1	6.4

Let us look at this result using statistical inference. 48 out of 62 students said that the applied examples were required and only 14 students said it was not necessary.

$$p = 0.774$$

$$q = 0.226$$

That is 0.774 said applied examples are necessary and 0.226 said it is not necessary. Now, with 95% confidence the idea of the student's population about the necessity of the applied examples for Math I, is as follows:

$$P - Z_{\frac{\alpha}{2}} \sqrt{\frac{pq}{n}} < \pi < P + Z_{\frac{\alpha}{2}} \sqrt{\frac{pq}{n}}$$

$$0.675 < \pi < 0.878$$

In which, π is the population's ratio. This means that with 95% confidence, the opinion of the population is between 67.5% to 87.8%, which shows the significant necessity of presenting applied examples in the class.

Question (5): How much is necessary to present applied examples in Math II?

The answers of the students to this question fall in 5 groups as follows:

Answer	Much More Required	More Required	It was Adequate	Less Required	Not Required At All
Frequency	22	18	16	6	0
Per Cent	35.5	29	25.8	9.7	0

56 out of 62 students said that the applied examples were adequate or they were needed

more. It means 90.3% agree that applied examples are required and only 9.7% said that it was not necessary. If we study the opinion of the population from the above relation, the result would be:

$$0.830 < \pi < 0.976$$

This means 83% to 97.6% of the population with confidence of 95% agrees that applied examples are necessary.

The results of the questionnaire and the extracted per cents show that the students accept the above teaching method and like it.

6) Comparison of the Teaching Methods

Teaching methods in universities other than Tehran and Shahid Beheshti Universities are different from my teaching method. Also in those universities most lectures are not familiar with economics. All the graduates of the universities must take the entrance examination to continue graduate studies at master level. The courses of the entrance examination are Micro and Macro Economics, Statistics, Mathematics, and English. Table below shows the number of students from each university who took this exam and their average in Mathematics in 2005. ^[9]

No	University	Participants	Average	Standard Deviation
1	Shahid Beheshti	202	37.98	24.78
2	Tehran	247	39.42	23.98
3	Shiraz	119	37.16	19.66
4	Isfahan	185	35.79	35.79
5	Allameh	399	31.67	31.67
6	Ferdousi Mashhad	163	29.53	29.53

The above table shows that the students of Tehran and Shahid Beheshti universities are more successful in the mathematical examinations than the students of the other universities. We divided these universities into two groups and compared their averages. In the first group we put Tehran University and Shahid Beheshti University, and the other universities form the second group. We use the test of hypothesis and show that there is a significant difference between the averages of these two groups of universities.

Universities	Average	N - number of students	S -Standard Deviation
Tehran and Shahid Beheshti	38.77	449	24
Shiraz, Isfahan, Allameh, and Ferdousi Mashhad	33.9	866	25

To test the hypothesis, if μ_1 is the average of Tehran and Shahid Beheshti universities students, and μ_2 is the average of the students of the other universities, we have:

$$H_0 : \mu_1 \geq \mu_2$$

$$H_1 : \mu_1 < \mu_2$$

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{38/77 - 33/90}{\sqrt{\frac{576}{449} + \frac{625}{866}}} = 2/05$$

Using test 1, $Z > 1/96$ is the range of $\alpha = 0.05$, so with confidence of 95%, $\mu_1 > \mu_2$. This concludes that, this teaching method is more successful than the other methods for the mathematical examinations.

5) Conclusion

As we show, non mathematics students are more interested in the courses other than mathematics. So we have to clarify the importance of mathematics for them. Therefore, the courses of mathematics have to be planed regarding the requirements of these branches. It would be more interested for the students if the presentation of each section includes the applied examples of mathematics in those areas. If the lecturer cannot give applied examples, then, s/he should encourage the students to find applied examples of mathematics in their other courses and represents it in the class. This will increase the interests of students in the mathematics.

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